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COMPUTER SCIENCE DEPARTMENT

Computer Networking & Data Communication

Finals Report

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Section: BSSE 5A

Network Topology Simulation (Packet Tracer)

Introduction

This project presents a comprehensive simulation of a Star/Ring topology network designed and implemented using Cisco Packet Tracer. The star topology, chosen for its centralized structure and ease of management, connects a hub, server, and five PCs to simulate real-world network functionality.

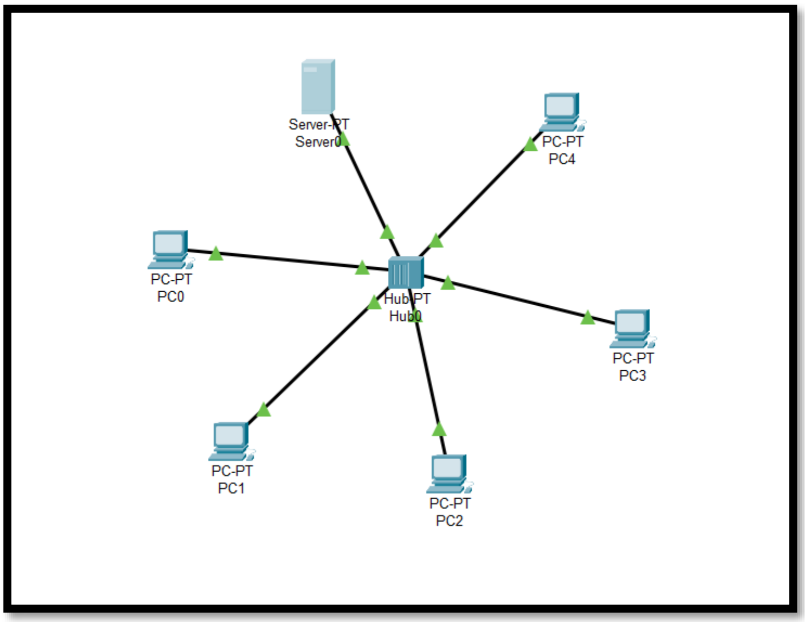
The simulation incorporates a variety of advanced networking features to demonstrate the configuration, operation, and analysis of key network protocols and services. These include:

1. **IP (Internet Protocol)** analysis and management using Wireshark to monitor data packet flow and ensure efficient addressing.
2. **ICMP (Internet Control Message Protocol)** testing and monitoring using Wireshark for efficient troubleshooting and error reporting.
3. **SSL (Secure Sockets Layer)** analysis using Wireshark to demonstrate secure data transmission across the network.
4. **Ethernet Protocol** and **ARP (Address Resolution Protocol)** simulation using Wireshark to analyze MAC-to-IP resolution and Ethernet communication.
5. **VLAN (Virtual Local Area Network)** configuration in Packet Tracer to segregate network traffic and enhance security.
6. **DHCP (Dynamic Host Configuration Protocol)** configuration for automatic IP address assignment, implemented and analyzed using both Packet Tracer and Wireshark.
7. **Dynamic Routing (OSPF)** OSPF is a dynamic routing protocol that ensures efficient route selection and network reliability between routers.
8. **UDP** is a connectionless protocol that enables fast data transmission with minimal overhead, ideal for real-time applications.

NOTE:

This simulation not only demonstrates the versatility and scalability of a star topology but also showcases the integration of modern networking protocols and tools in a controlled environment. The project provides a detailed insight into the workings of key network features, highlighting their roles in establishing a robust and functional network.

Star Topology



Creating a Star Topology in Cisco Packet Tracer

A star topology was designed using 1 hub, 1 server, and 5 PCs in Cisco Packet Tracer. The process was carried out in the following steps:

Step 1: Deploying the Devices

Selecting the required devices from the Network Devices and End Devices sections. A 10-port hub was placed at the center of the workspace to act as the central connecting point. A server and five PCs were then added and positioned around the hub for clarity and accessibility.

Step 2: Establishing Connections

Connecting the hub to each device using Copper Straight-Through cables. Each PC's FastEthernet port was connected to an available port on the hub. Similarly, the server's FastEthernet interface was linked to the hub, completing the physical connections.

Step 3: Configuring IP Addresses

Assigning unique IP addresses to enable communication across the devices:

- The server was configured with an IP address of 192.168.0.
- Each PC was assigned sequential IP addresses, starting from 192.168.0.2 to 192.168.0.6, all using the same subnet mask.

Step 4: Verifying Connectivity

Using the Add Simple PDU tool to simulate data transmission between the server and the PCs. Additionally, performing ping tests from the Command Prompt of each device confirmed seamless communication across the network.

Step 5: Finalizing and Documenting

Saving the project as a .pkt file for future use. The network design, including device roles, IP configurations, and topology structure, was documented to ensure clarity and reproducibility.

Internet Protocol (IP) Traffic

Steps for Packet Tracer

Step 1: Set Up IP Configuration:

- IP addresses are assigned to all PCs and servers in the star topology.

Step 2: Generate IP Traffic:

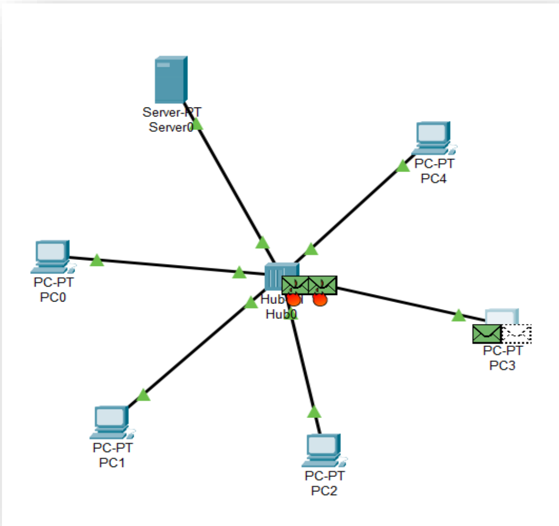
- The Command Prompt is used on a PC to send packets:
- ping 192.168.1.1
- This simulates ICMP traffic using IP.

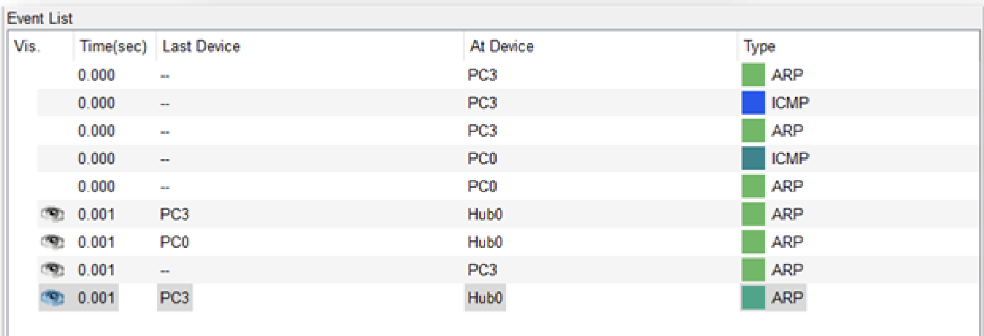
Step 3: Observe the Traffic:

- Simulation Mode is activated in Packet Tracer.
- The flow of packets between devices is observed in real-time.
- The Event List is used to monitor each packet's journey through the network.

Step 4: Verify Communication:

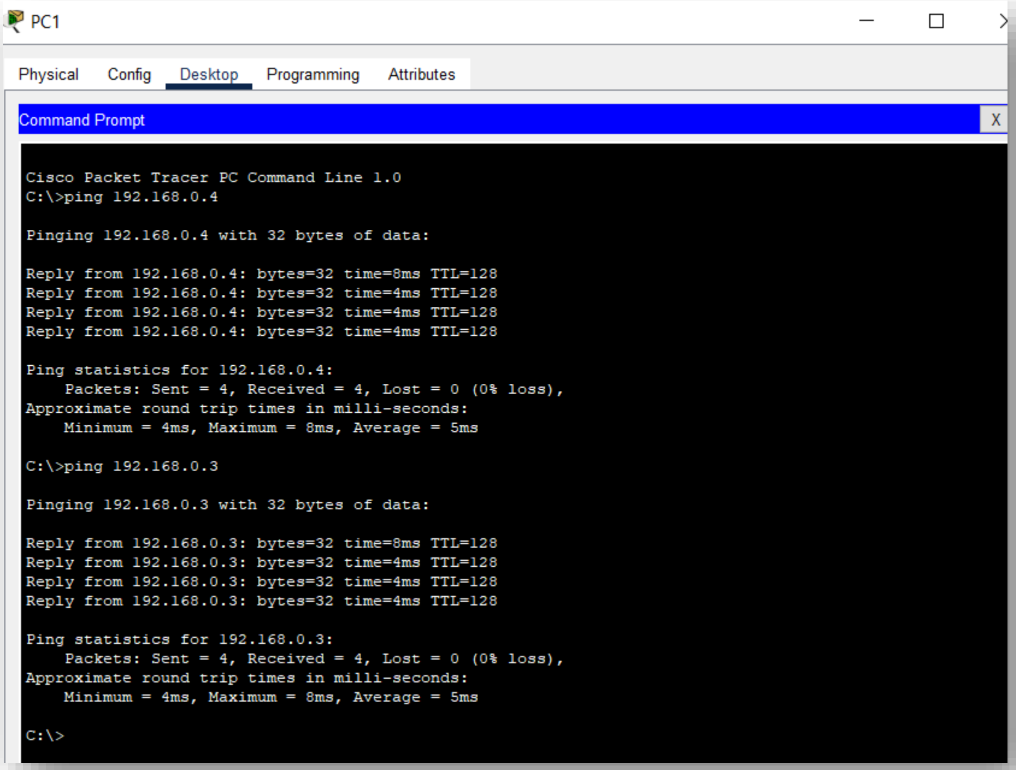
- It is ensured that packets reach their destination without errors.
- If packets fail, troubleshooting is performed by checking IP configurations, cable connections, or device status.




Event List

Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC3	ARP
	0.000	--	PC3	ICMP
	0.000	--	PC3	ARP
	0.000	--	PC0	ICMP
	0.000	--	PC0	ARP
	0.001	PC3	Hub0	ARP
	0.001	PC0	Hub0	ARP
	0.001	--	PC3	ARP
	0.001	PC3	Hub0	ARP

Pinging other pc's in star topology



Steps for Wireshark

Step 1: Generate IP Traffic:

- The ping command or other network activity is used on a PC to generate traffic. Example for ping 192.168.1.1
- This generates ICMP packets encapsulated within IP headers.

Step 2: Capture Traffic:

- Wireshark is opened on the same device or another device connected to the network.
- A packet capture is started on the network interface connected to the hub.

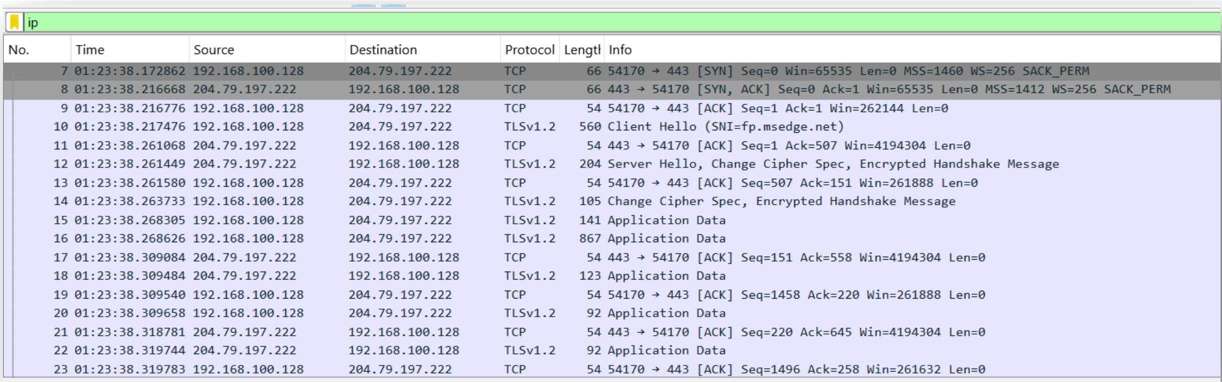
Step 3: Filter for IP Packets:

- The following Wireshark filter is applied: ip
- This isolates all IP packets in the captured traffic.

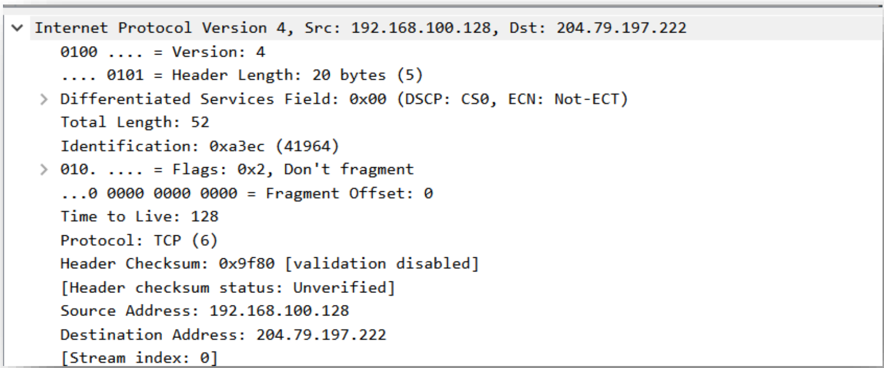
Step 4: Analyze the Captured Traffic:

- **Source and Destination IPs** are examined to identify the communication endpoints.
- **TTL (Time to Live)** is checked to observe how many hops packets take before expiration.
- The Protocol is inspected for encapsulated data like **TCP, UDP, or ICMP**.
- Observations are made on the successful transmission or loss of packets.
- This version describes the processes without directly addressing the user, maintaining a third-person narrative throughout.

Filtering in WireShark



Packet Details Pane



ICMP(Internet Control Message Protocol)

Steps for Packet Tracer

Step 1: Setup the Network Configuration:

- A star topology was created in Cisco Packet Tracer, consisting of one hub connected to five PCs and one server using copper straight-through cables.
- Static IP addresses were assigned to all devices to ensure proper connectivity.

Step 2: Enable Device Communication:

- Each device was verified for connectivity by ensuring the network interface cards (NICs) were powered on and connected to the hub.

Step 3: Generate ICMP Traffic:

- From one of the PCs, the ping command was executed in the Command Prompt to test connectivity to another device in the topology.
  Example: ping 192.168.1.1

- This command generated ICMP Echo Request packets, which were sent to the destination IP address.

Step 4: Monitor ICMP Traffic:

- Packet Tracer's Simulation Mode was activated to observe the flow of ICMP packets between devices.
- The Event List displayed the journey of packets, showing Echo Request and Echo Reply messages between the source and destination.

Step 5: Verify Communication:

- Successful communication was confirmed by checking the ping responses.
- If the destination device was reachable, replies were displayed in the Command Prompt, indicating successful ICMP traffic.

ICMP traffic generated in packet tracer

Vis.	Time(sec)	Last Device	At Device	Type
	252.597	Hub0	PC4	ICMP
	252.597	Hub0	Server0	ICMP
	252.598	PC3	Hub0	ICMP
	252.599	Hub0	PC1	ICMP
	252.599	Hub0	PC2	ICMP
	252.599	Hub0	PC4	ICMP
	252.599	Hub0	Server0	ICMP
	252.599	Hub0	PC0	ICMP
	721.495	--	PC0	ICMP
	721.496	PC0	Hub0	ICMP
	721.497	Hub0	PC1	ICMP
	721.497	Hub0	PC2	ICMP
	721.497	Hub0	PC3	ICMP
	721.497	Hub0	PC4	ICMP
	721.497	Hub0	Server0	ICMP
	721.498	PC3	Hub0	ICMP

Pinging between Pc's to verify stable connection

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.0.3

Pinging 192.168.0.3 with 32 bytes of data:

Reply from 192.168.0.3: bytes=32 time=1ms TTL=128
Reply from 192.168.0.3: bytes=32 time=1ms TTL=128
Reply from 192.168.0.3: bytes=32 time=1ms TTL=128
Reply from 192.168.0.3: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.0.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>ping 192.168.0.5

Pinging 192.168.0.5 with 32 bytes of data:

Reply from 192.168.0.5: bytes=32 time<1ms TTL=128
Reply from 192.168.0.5: bytes=32 time=1ms TTL=128
Reply from 192.168.0.5: bytes=32 time<1ms TTL=128
Reply from 192.168.0.5: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.0.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Steps for WireShark

Step 1: Filter ICMP Packets in Wireshark:

- During the packet capture, the filter icmp was applied in Wireshark to isolate ICMP traffic from other network packets.
- This filter displayed only the Echo Request and Echo Reply packets.

Step 2: Analyze Captured ICMP Traffic:

- The captured packets were analyzed in Wireshark's Packet List pane.
- Detailed information about the packets, such as source and destination IP addresses, ICMP types (Request or Reply), and TTL values, was reviewed in the Packet Details pane.

Step 3: Verify Communication:

- Successful ICMP communication was verified by observing the presence of both Echo Request and Echo Reply packets in Wireshark.
- Any packet loss or delays were noted during the analysis.

Filtering ICMP Traffic

icmp

No.	Time	Source	Destination	Protocol	Length	Info
7203	16:50:12.826223	192.168.100.128	40.99.43.50	ICMP	44	Echo (ping) request id=0x0001, seq=12/3072, ttl=30 (reply in 7222)
7222	16:50:13.008672	40.99.43.50	192.168.100.128	ICMP	44	Echo (ping) reply id=0x0001, seq=12/3072, ttl=235 (request in 7203)
7334	16:50:14.745881	192.168.100.128	52.98.84.114	ICMP	44	Echo (ping) request id=0x0001, seq=13/3328, ttl=30 (reply in 7335)
7335	16:50:14.886732	52.98.84.114	192.168.100.128	ICMP	44	Echo (ping) reply id=0x0001, seq=13/3328, ttl=238 (request in 7334)
7554	17:30:23.527723	192.168.100.128	40.99.68.34	ICMP	44	Echo (ping) request id=0x0001, seq=14/3584, ttl=30 (reply in 7558)
7558	17:30:23.573062	40.99.68.34	192.168.100.128	ICMP	44	Echo (ping) reply id=0x0001, seq=14/3584, ttl=233 (request in 7554)
7638	17:30:23.900139	192.168.100.128	52.98.32.2	ICMP	44	Echo (ping) request id=0x0001, seq=15/3840, ttl=30 (reply in 7674)
7674	17:30:23.945539	52.98.32.2	192.168.100.128	ICMP	44	Echo (ping) reply id=0x0001, seq=15/3840, ttl=233 (request in 7638)

Echo (ping) request

```
> Frame 7203: 44 bytes on wire (352 bits), 44 bytes captured (352 bits) on interface \Device\NPF_{FC924494-1E54-434E-865E-D49D262A3739}, id 0
> Ethernet II, Src: Intel_da:8d:10 (70:1c:e7:da:8d:10), Dst: HuaweiTechno_ac:cc:dd (2c:55:d3:ac:cc:dd)
> Internet Protocol Version 4, Src: 192.168.100.128, Dst: 40.99.43.50
v Internet Control Message Protocol
    Type: 8 (Echo (ping) request)
    Code: 0
    Checksum: 0xf7f2 [correct]
    [Checksum Status: Good]
    Identifier (BE): 1 (0x0001)
    Identifier (LE): 256 (0x0100)
    Sequence Number (BE): 12 (0x000c)
    Sequence Number (LE): 3072 (0x0c00)
    [Response frame: 7222]
  v Data (2 bytes)
      Data: 0000
      [Length: 2]
```

Echo (ping) reply

```
> Frame 7222: 44 bytes on wire (352 bits), 44 bytes captured (352 bits) on interface \Device\NPF_{FC924494-1E54-434E-865E-D49D262A3739}, id 0
> Ethernet II, Src: HuaweiTechno_ac:cc:dd (2c:55:d3:ac:cc:dd), Dst: Intel_da:8d:10 (70:1c:e7:da:8d:10)
> Internet Protocol Version 4, Src: 40.99.43.50, Dst: 192.168.100.128
v Internet Control Message Protocol
    Type: 0 (Echo (ping) reply)
    Code: 0
    Checksum: 0xfff2 [correct]
    [Checksum Status: Good]
    Identifier (BE): 1 (0x0001)
    Identifier (LE): 256 (0x0100)
    Sequence Number (BE): 12 (0x000c)
    Sequence Number (LE): 3072 (0x0c00)
    [Request frame: 7203]
    [Response time: 182.449 ms]
  v Data (2 bytes)
      Data: 0000
      [Length: 2]
```

Secure Socket Layer Using WireShark

Step 1: Filter SSL Traffic in Wireshark:

- During the capture, the filter ssl or tls was applied in Wireshark to isolate SSL/TLS packets.
- The captured packets displayed the handshake process, encrypted data, and other SSL-related details.

Step 2: Analyze Captured SSL Traffic:

- The Packet List pane in Wireshark displayed the SSL handshake, including:
  - **Client Hello:** Initiation of the SSL connection by the client.
  - **Server Hello:** Response from the server with SSL parameters.
  - **Certificate Exchange:** Server's SSL certificate for authentication.
  - **Encrypted Handshake:** Transition to encrypted communication.
- Subsequent packets showed encrypted application data transferred between the client and server.

Step 3: Verify Secure Communication:

- Successful SSL communication was verified by observing the completion of the handshake and the encrypted data transfer.
- Any issues, such as handshake failures, were resolved by reviewing the SSL configuration on the server.

Filtering SSL Traffic

ssl || tls

No.	Time	Source	Destination	Protocol	Lengtl	Info
27	13:30:59.187490	2600:1f18:24e6:b900…	2407:d000:a:2f3c:d4…	TLSv1.2	113	Application Data
29	13:30:59.394614	2600:1f18:24e6:b900…	2407:d000:a:2f3c:d4…	TLSv1.2	98	Application Data
35	13:31:01.312335	2407:d000:a:2f3c:d4…	2a03:2880:f34a:120:…	TLSv1.2	144	Application Data
37	13:31:01.530768	2a03:2880:f34a:120:…	2407:d000:a:2f3c:d4…	TLSv1.2	146	Application Data
63	13:31:26.320234	2407:d000:a:2f3c:d4…	2a03:2880:f34a:120:…	TLSv1.2	144	Application Data
65	13:31:26.824029	2a03:2880:f34a:120:…	2407:d000:a:2f3c:d4…	TLSv1.2	146	Application Data
108	13:31:46.218911	2407:d000:a:2f3c:d4…	2620:1ec:42::132	TLSv1.2	273	Client Hello (SNI=ecs.office.com)
114	13:31:46.367756	2620:1ec:42::132	2407:d000:a:2f3c:d4…	TLSv1.2	408	Server Hello, Certificate, Certificate Status, Server Key Exchange, Server Hello Done
116	13:31:46.375523	2407:d000:a:2f3c:d4…	2620:1ec:42::132	TLSv1.2	232	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
118	13:31:46.525519	2620:1ec:42::132	2407:d000:a:2f3c:d4…	TLSv1.2	416	New Session Ticket, Change Cipher Spec, Encrypted Handshake Message
119	13:31:46.525519	2620:1ec:42::132	2407:d000:a:2f3c:d4…	TLSv1.2	143	Application Data

Client Hello

```
Transport Layer Security
  TLSv1.2 Record Layer: Handshake Protocol: Client Hello
    Content Type: Handshake (22)
    Version: TLS 1.2 (0x0303)
    Length: 194
    Handshake Protocol: Client Hello
      Handshake Type: Client Hello (1)
      Length: 190
      Version: TLS 1.2 (0x0303)
      Random: 676fb77286bf6d7676afdcd51e686eebaf9525eef425c7ad9b4dcb3dd5b03bef
      Session ID Length: 0
      Cipher Suites Length: 38
      Cipher Suites (19 suites)
      Compression Methods Length: 1
      Compression Methods (1 method)
      Extensions Length: 111
      Extension: server_name (len=19) name=ecs.office.com
      Extension: status_request (len=5)
      Extension: supported_groups (len=8)
      Extension: ec_point_formats (len=2)
      Extension: signature_algorithms (len=26)
      Extension: session_ticket (len=0)
      Extension: application_layer_protocol_negotiation (len=14)
      Extension: extended_master_secret (len=0)
      Extension: renegotiation_info (len=1)
      [JA4: t12d1909h2_d83cc789557e_7af1ed941c26]
      [JA4_r: t12d1909h2_000a,002f,0035,003c,003d,009c,009d,c009,c00a,c013,c014,c023,c024,c027,c028,c02b,c02c,c02f,c030_0005,000a,000b,000d,0017,0023,f
      [JA3 Fullstring: 771,49196-49195-49200-49199-49188-49187-49192-49191-49162-49161-49172-49171-157-156-61-60-53-47-10,0-5-10-11-13-35-16-23-65281,2
      [JA3: 28a2c9bd18a11de089ef85a160da29e4]
```

Server name

```
Extension: server_name (len=19) name=ecs.office.com
  Type: server_name (0)
  Length: 19
  Server Name Indication extension
    Server Name list length: 17
    Server Name Type: host_name (0)
    Server Name length: 14
    Server Name: ecs.office.com
```

Server Hello, Certificate

```
114 13:31:46.367756 2620:1ec:42::132     2407:d000:a:2f3c:d4… TLSv1.2   408 Server Hello, Certificate, Certificate Status, Server Key Exchange, Server Hello Done

Frame 114: 408 bytes on wire (3264 bits), 408 bytes captured (3264 bits) on interface \Device\NPF_{FC924494-1E54-434E-865E-D49D262A3739}, id 0
Ethernet II, Src: HuaweiTechno_ac:cc:dd (2c:55:d3:ac:cc:dd), Dst: Intel_da:8d:10 (70:1c:e7:da:8d:10)
Internet Protocol Version 6, Src: 2620:1ec:42::132, Dst: 2407:d000:a:2f3c:d4d5:15d1:fe51:328f
Transmission Control Protocol, Src Port: 443, Dst Port: 60765, Seq: 5649, Ack: 200, Len: 334
[5 Reassembled TCP Segments (5982 bytes): #110(1412), #111(1412), #112(1412), #113(1412), #114(334)]
Transport Layer Security
  TLSv1.2 Record Layer: Handshake Protocol: Multiple Handshake Messages
    Content Type: Handshake (22)
    Version: TLS 1.2 (0x0303)
    Length: 5977
    Handshake Protocol: Server Hello
    Handshake Protocol: Certificate
    Handshake Protocol: Certificate Status
    Handshake Protocol: Server Key Exchange
    Handshake Protocol: Server Hello Done
```

```
0020  27 0f 56 44 4f
0030  03 a6 41 6b 58
0040  ae 09 10 fb 29
0050  1e 00 05 00 00
0060  68 32 00 17 00
0070  00 0e 30 00 0e
0080  5b a0 03 02 01
0090  e9 de 01 da f3
00a0  86 48 86 f7 0d
00b0  06 03 55 04 06
00c0  04 0a 13 15 4c
00d0  72 70 6f 72 61
00e0  04 03 13 25 4c
00f0  75 72 65 20 52
0100  69 6e 67 20 43
0110  32 31 35 32 31
0120  31 33 32 31 32
0130  03 55 04 06 13
```

Certificate Details

```
> Handshake Protocol: Server Hello
v Handshake Protocol: Certificate
    Handshake Type: Certificate (11)
    Length: 3632
    Certificates Length: 3629
  > Certificates (3629 bytes)
v Handshake Protocol: Certificate Status
    Handshake Type: Certificate Status (22)
    Length: 1862
    Certificate Status Type: OCSP (1)
    OCSP Response Length: 1858
  > OCSP Response
v Handshake Protocol: Server Key Exchange
    Handshake Type: Server Key Exchange (12)
    Length: 361
  > EC Diffie-Hellman Server Params
v Handshake Protocol: Server Hello Done
    Handshake Type: Server Hello Done (14)
    Length: 0
```

Encrypted Data

```
> Frame 119: 143 bytes on wire (1144 bits), 143 bytes captured (1144 bits) on interface \Device\NPF_{FC924494-1E54-434E-865E-D49D262A3739}, id 0
> Ethernet II, Src: HuaweiTechno_ac:cc:dd (2c:55:d3:ac:cc:dd), Dst: Intel_da:8d:10 (70:1c:e7:da:8d:10)
> Internet Protocol Version 6, Src: 2620:1ec:42::132, Dst: 2407:d000:a:2f3c:d4d5:15d1:fe51:328f
> Transmission Control Protocol, Src Port: 443, Dst Port: 60765, Seq: 6325, Ack: 358, Len: 69
v Transport Layer Security
  v TLSv1.2 Record Layer: Application Data Protocol: HyperText Transfer Protocol 2
      Content Type: Application Data (23)
      Version: TLS 1.2 (0x0303)
      Length: 64
      Encrypted Application Data: 0000000000000001d4040bfbc191d242613458dac030c9e537c3f4a26c3314b544da7ee0396bb3a24c571927396822e762e5527758f3dbedcf7bbd0a7ce89d65
      [Application Data Protocol: HyperText Transfer Protocol 2]
```

(ARP)Address Resolution protocol using Wireshark

Step 1: Filter for ARP Packets:

- During the capture, the filter arp was applied in Wireshark to display only ARP traffic.
- The captured packets included:
  - **ARP Request:** Sent by the PC to discover the MAC address of the target device.
  - **ARP Reply:** Sent by the target device containing its MAC address.

Step 2: Analyze ARP Traffic in Wireshark:

- The captured ARP packets were reviewed to understand their contents:
  - **Sender MAC Address:** The MAC address of the machine sending the ARP request or reply.
  - **Sender IP Address:** The IP address of the machine sending the ARP request or reply.
  - **Target MAC Address:** The MAC address of the machine that the sender is trying to reach.
  - **Target IP Address:** The IP address the sender is trying to resolve to a MAC address.
- The ARP process was verified by observing successful replies to the requests

Filtering ARP Traffic

arp

No.	Time	Source	Destination	Protocol	Length	Info
289	01:24:14.218778	Intel_da:8d:10	HuaweiTechno_ac:cc:…	ARP	42	Who has 192.168.100.1? Tell 192.168.100.128
290	01:24:14.220111	HuaweiTechno_ac:cc:…	Intel_da:8d:10	ARP	42	192.168.100.1 is at 2c:55:d3:ac:cc:dd
1653	01:25:07.728274	Intel_da:8d:10	HuaweiTechno_ac:cc:…	ARP	42	Who has 192.168.100.1? Tell 192.168.100.128
1654	01:25:07.729359	HuaweiTechno_ac:cc:…	Intel_da:8d:10	ARP	42	192.168.100.1 is at 2c:55:d3:ac:cc:dd
1808	01:25:37.221882	Intel_da:8d:10	HuaweiTechno_ac:cc:…	ARP	42	Who has 192.168.100.1? Tell 192.168.100.128
1809	01:25:37.223234	HuaweiTechno_ac:cc:…	Intel_da:8d:10	ARP	42	192.168.100.1 is at 2c:55:d3:ac:cc:dd
2022	01:26:02.223439	Intel_da:8d:10	HuaweiTechno_ac:cc:…	ARP	42	Who has 192.168.100.1? Tell 192.168.100.128
2023	01:26:02.224483	HuaweiTechno_ac:cc:…	Intel_da:8d:10	ARP	42	192.168.100.1 is at 2c:55:d3:ac:cc:dd
2232	01:26:14.723934	Intel_da:8d:10	HuaweiTechno_ac:cc:…	ARP	42	Who has 192.168.100.1? Tell 192.168.100.128
2233	01:26:14.725118	HuaweiTechno_ac:cc:…	Intel_da:8d:10	ARP	42	192.168.100.1 is at 2c:55:d3:ac:cc:dd
2239	01:26:22.498924	HuaweiDevice_48:05:…	Broadcast	ARP	42	Who has 192.168.100.1? Tell 192.168.100.32
2427	01:27:25.719603	HuaweiDevice_48:03:…	Broadcast	ARP	42	Who has 192.168.100.1? Tell 192.168.100.31
2438	01:27:34.228636	Intel_da:8d:10	HuaweiTechno_ac:cc:…	ARP	42	Who has 192.168.100.1? Tell 192.168.100.128
2439	01:27:34.229868	HuaweiTechno_ac:cc:…	Intel_da:8d:10	ARP	42	192.168.100.1 is at 2c:55:d3:ac:cc:dd
2486	01:28:05.517641	HuaweiTechno_ac:cc:…	Intel_da:8d:10	ARP	42	Who has 192.168.100.128? Tell 192.168.100.1
2487	01:28:05.517687	Intel_da:8d:10	HuaweiTechno_ac:cc:…	ARP	42	192.168.100.128 is at 70:1c:e7:da:8d:10
2534	01:28:11.216752	Intel_da:8d:10	HuaweiTechno_ac:cc:…	ARP	42	Who has 192.168.100.1? Tell 192.168.100.128
2535	01:28:11.219136	HuaweiTechno_ac:cc:…	Intel_da:8d:10	ARP	42	192.168.100.1 is at 2c:55:d3:ac:cc:dd
2748	01:29:34.219218	Intel_da:8d:10	HuaweiTechno_ac:cc:…	ARP	42	Who has 192.168.100.1? Tell 192.168.100.128
2749	01:29:34.220602	HuaweiTechno_ac:cc:…	Intel_da:8d:10	ARP	42	192.168.100.1 is at 2c:55:d3:ac:cc:dd

Request

```
> Frame 289: 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF_{FC924494-1E54-434E-865E-D49D262A3739}, id 0
> Ethernet II, Src: Intel_da:8d:10 (70:1c:e7:da:8d:10), Dst: HuaweiTechno_ac:cc:dd (2c:55:d3:ac:cc:dd)
v Address Resolution Protocol (request)
    Hardware type: Ethernet (1)
    Protocol type: IPv4 (0x0800)
    Hardware size: 6
    Protocol size: 4
    Opcode: request (1)
    Sender MAC address: Intel_da:8d:10 (70:1c:e7:da:8d:10)
    Sender IP address: 192.168.100.128
    Target MAC address: HuaweiTechno_ac:cc:dd (2c:55:d3:ac:cc:dd)
    Target IP address: 192.168.100.1
```

Reply

```
> Frame 290: 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF_{FC924494-1E54-434E-865E-D49D262A3739}, id 0
> Ethernet II, Src: HuaweiTechno_ac:cc:dd (2c:55:d3:ac:cc:dd), Dst: Intel_da:8d:10 (70:1c:e7:da:8d:10)
v Address Resolution Protocol (reply)
    Hardware type: Ethernet (1)
    Protocol type: IPv4 (0x0800)
    Hardware size: 6
    Protocol size: 4
    Opcode: reply (2)
    Sender MAC address: HuaweiTechno_ac:cc:dd (2c:55:d3:ac:cc:dd)
    Sender IP address: 192.168.100.1
    Target MAC address: Intel_da:8d:10 (70:1c:e7:da:8d:10)
    Target IP address: 192.168.100.128
```

Ethernet Using Wireshark

Step 1: Filtering Ethernet Traffic

In Wireshark, the focus was on Ethernet packets by applying filters to isolate the relevant traffic. A filter query, eth, was utilized to display only Ethernet-related traffic, making the analysis process more straightforward and efficient.

Step 2: Analysis of Packet Details

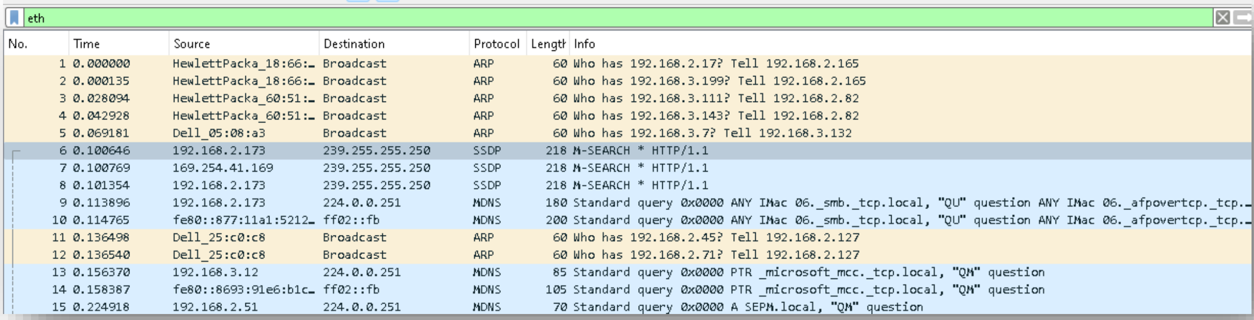
Detailed packet information provided by Wireshark was carefully examined. Key aspects of the analysis included:

- **Source and Destination MAC Addresses**: Used to identify the devices involved in communication within the star topology.
- **Packet Type**: Including protocols such as ARP (Address Resolution Protocol) or IPv4/IPv6 traffic encapsulated within Ethernet frames.
- **Payload Data**: Observing the encapsulated data transmitted across the network, which provided insights into the communication structure and content.

Step 3: Capturing Screenshots of Filtered Traffic

To visually document the analysis, screenshots of the filtered Ethernet traffic were captured. These screenshots served as concrete evidence of network communication and were incorporated into the project as a reference for further study and reporting.

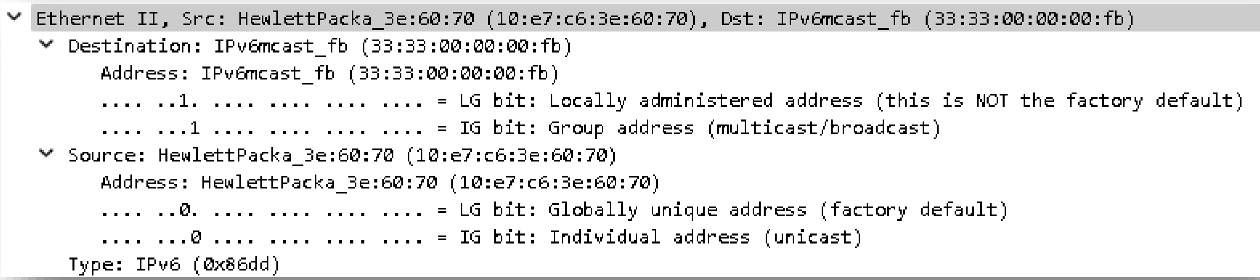
Ethernet Traffic Capture



eth

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	HewlettPacka_18:66:…	Broadcast	ARP	60	Who has 192.168.2.17? Tell 192.168.2.165
2	0.000135	HewlettPacka_18:66:…	Broadcast	ARP	60	Who has 192.168.3.199? Tell 192.168.2.165
3	0.028094	HewlettPacka_60:51:…	Broadcast	ARP	60	Who has 192.168.3.111? Tell 192.168.2.82
4	0.042928	HewlettPacka_60:51:…	Broadcast	ARP	60	Who has 192.168.3.143? Tell 192.168.2.82
5	0.069181	Dell_05:08:a3	Broadcast	ARP	60	Who has 192.168.3.7? Tell 192.168.3.132
6	0.100646	192.168.2.173	239.255.255.250	SSDP	218	M-SEARCH * HTTP/1.1
7	0.100769	169.254.41.169	239.255.255.250	SSDP	218	M-SEARCH * HTTP/1.1
8	0.101354	192.168.2.173	239.255.255.250	SSDP	218	M-SEARCH * HTTP/1.1
9	0.113896	192.168.2.173	224.0.0.251	MDNS	180	Standard query 0x0000 ANY IMac 06._smb._tcp.local, "QU" question ANY IMac 06._afpovertcp._tcp.…
10	0.114765	fe80::877:11a1:5212…	ff02::fb	MDNS	200	Standard query 0x0000 ANY IMac 06._smb._tcp.local, "QU" question ANY IMac 06._afpovertcp._tcp.…
11	0.136498	Dell_25:c0:c8	Broadcast	ARP	60	Who has 192.168.2.45? Tell 192.168.2.127
12	0.136540	Dell_25:c0:c8	Broadcast	ARP	60	Who has 192.168.2.71? Tell 192.168.2.127
13	0.156370	192.168.3.12	224.0.0.251	MDNS	85	Standard query 0x0000 PTR _microsoft_mcc._tcp.local, "QM" question
14	0.158387	fe80::8693:91e6:b1c…	ff02::fb	MDNS	105	Standard query 0x0000 PTR _microsoft_mcc._tcp.local, "QM" question
15	0.224918	192.168.2.51	224.0.0.251	MDNS	70	Standard query 0x0000 A SEPM.local, "QM" question

Packet Details



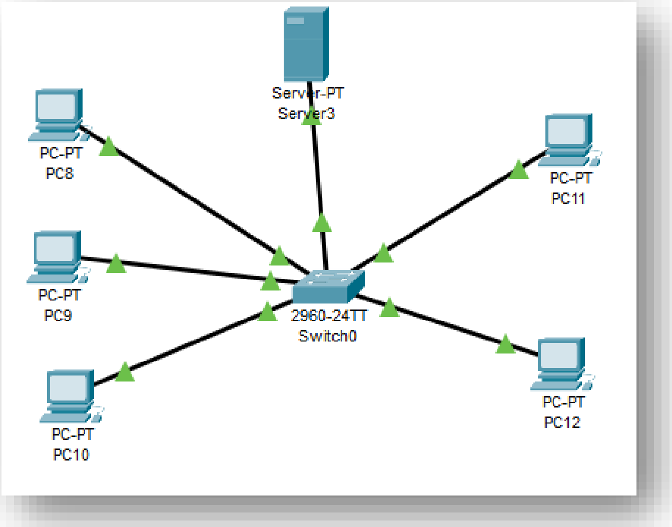
```
Ethernet II, Src: HewlettPacka_3e:60:70 (10:e7:c6:3e:60:70), Dst: IPv6mcast_fb (33:33:00:00:00:fb)
  Destination: IPv6mcast_fb (33:33:00:00:00:fb)
    Address: IPv6mcast_fb (33:33:00:00:00:fb)
    .... ..1. .... .... .... .... = LG bit: Locally administered address (this is NOT the factory default)
    .... ...1 .... .... .... .... = IG bit: Group address (multicast/broadcast)
  Source: HewlettPacka_3e:60:70 (10:e7:c6:3e:60:70)
    Address: HewlettPacka_3e:60:70 (10:e7:c6:3e:60:70)
    .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
    .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
  Type: IPv6 (0x86dd)
```

Configuring VLAN Using Packet Tracer

Steps to Configure VLANs:

Step 1: Replace the Hub with a Switch

- Physical Layer: Replace the hub with a Layer 2 switch (e.g., 2960 Switch).
- Connect the switch to the router if the router is used for inter-VLAN routing or connects to the outside network.
- Connect the 4 PCs and server to the switch.



Step 2: Configure VLANs on the Switch

- **Enter Switch Configuration Mode**:
  - Click on the switch and enter configuration mode.
- **Create VLANs**:
  - Define two VLANs (you can define more depending on your needs). For example:
    - **VLAN 10**: For PC1 and PC2.
    - **VLAN 20**: For PC3 and PC4.

Step 3: Assign Ports to VLANs

- After creating the VLANs, you need to assign switch ports to the respective VLANs.
  - PC1 and PC2 will be on VLAN 10.
  - PC3 and PC4 will be on VLAN 20.
  - The **server** will be on **VLAN 30**.

Step 4: Configure Switch for Inter-VLAN Routing (if required)

If you want devices in different VLANs to communicate with each other, you need to configure Router-on-a-Stick (Inter-VLAN Routing). You will need to:

- **Create Subinterfaces on the Router**:
  - On the router, create subinterfaces for each VLAN (e.g., Gig0/1.10, Gig0/1.20, Gig0/1.30).
  - Assign IP addresses to each subinterface in their respective VLAN subnets.

Step 5: Configure PCs and Server with IP Addresses

- Assign **static IP addresses** to the PCs and server according to the VLAN they are in:
  - **PC1** and **PC2**: 192.168.10.2 and 192.168.10.3 (VLAN 10), with default gateway 192.168.10.1.
  - **PC3** and **PC4**: 192.168.20.2 and 192.168.20.3 (VLAN 20), with default gateway 192.168.20.1.
  - **Server**: 192.168.30.2 (VLAN 30), with default gateway 192.168.30.1.

Step 6: Verify VLAN Configuration

- **Ping Test**: Check if devices within the same VLAN can communicate and verify that devices across VLANs can communicate (if Inter-VLAN routing is configured).
  - For example, try pinging from PC1 to PC2 (same VLAN), and from PC1 to PC3 (different VLAN).
- On the switch, check the VLAN assignments:

- Switch# show vlan brief
- On the router, verify subinterfaces and IPs:
- Router# show ip interface brief

```
enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan10
                   ^
% Invalid input detected at '^' marker.

Switch(config)# vlan 10
Switch(config-vlan)# name VLAN10
Switch(config-vlan)# exit
Switch(config)# vlan 20
Switch(config-vlan)# name VLAN20
Switch(config-vlan)# exit
Switch(config)# vlan 30
Switch(config-vlan)# name VLAN30
Switch(config-vlan)# exit
Switch(config)# interface range fa0/1 - fa/2
                                          ^
% Invalid input detected at '^' marker.

Switch(config)# interface range fa0/1 - fa0/2
Switch(config-if-range)# switchport mode access
Switch(config-if-range)#exit
Switch(config)# interface fa0/5
Switch(config-if)# switchport mode access
Switch(config-if)# exit
Switch(config)# interface range fa0/3 - fa0/4
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport vlan 20
                                     ^
% Invalid input detected at '^' marker.

Switch(config-if-range)# switchport access vlan 20
Switch(config-if-range)# exit
Switch(config)# interface range fa0/1 - fa0/2
Switch(config-if-range)# switchport mode access
Switch(config-if-range)#switch port access vlan 10
                                     ^
% Invalid input detected at '^' marker.

Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#exit
Switch(config)#interface fa0/5
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#exit
Switch(config)#
```

```
Command Prompt                                                    X

Packet Tracer PC Command Line 1.0
C:\> ping 192.168.10.3

Pinging 192.168.10.3 with 32 bytes of data:

Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Dynamic Host Configuration Protocol

Steps for Packet Tracer

Step 1: Configure the Server as a DHCP Server

- The user selects the server and goes to the Services Tab.
- The user opens the DHCP service and turns it *ON*.
- The user creates a DHCP pool:
  - The pool name is entered.
  - The **Default Gateway** is set to 192.168.1.1.
  - The **Subnet Mask** is defined as 255.255.255.0.
  - The **Start IP Address** is specified as 192.168.1.2.
- The user saves the configuration.

Step 2: Connect Devices to the Network

- The user connects the server, PCs, and hub using straight-through cables.
- The user ensures the hub is powered on and functioning properly.

Step 3: Assign a Static IP to the Server

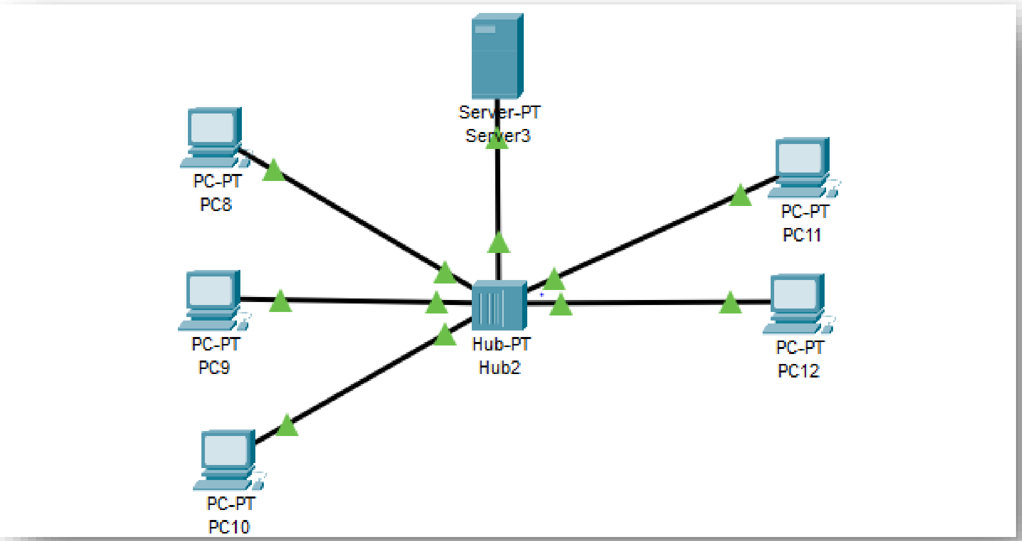
- The user assigns the server a static IP address within the same subnet as the DHCP pool:
  - IP Address: 192.168.1.2
  - Subnet Mask: 255.255.255.0
  - Default Gateway: 192.168.1.1

Step 4: Configure PCs to Use DHCP

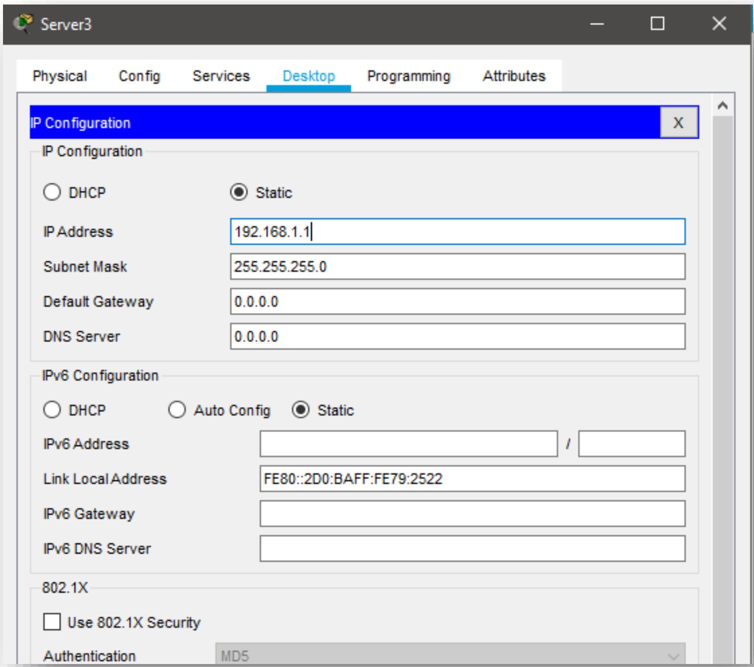
- For each PC:
  - The user goes to Desktop > IP Configuration.
  - The user sets the IP configuration to DHCP.

Step 5: Test DHCP and Network Connectivity

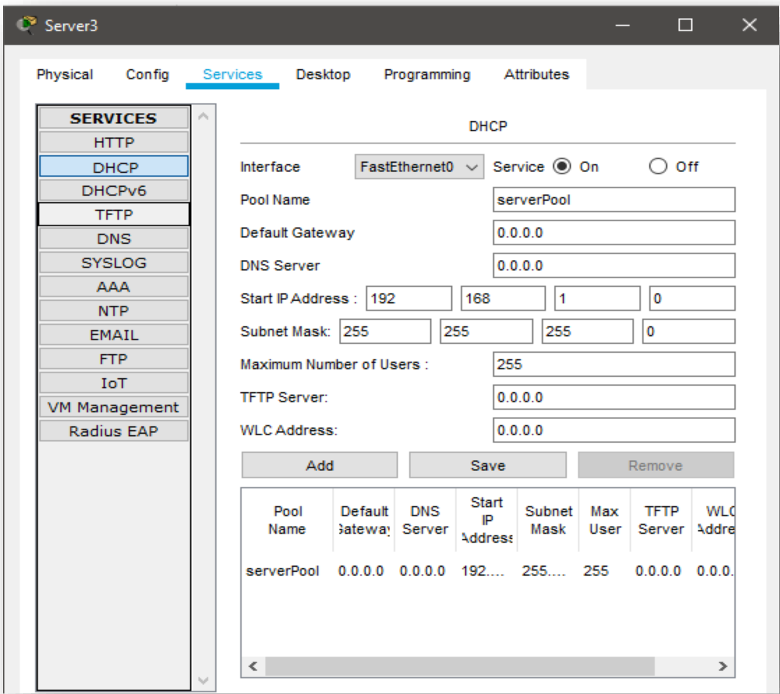
- The user uses the ipconfig command on each PC to verify the assigned IP address.
- The user tests communication by pinging the server from any PC: ping <Server_IP>
- The user ensures that all PCs receive IP addresses and can communicate with the server.



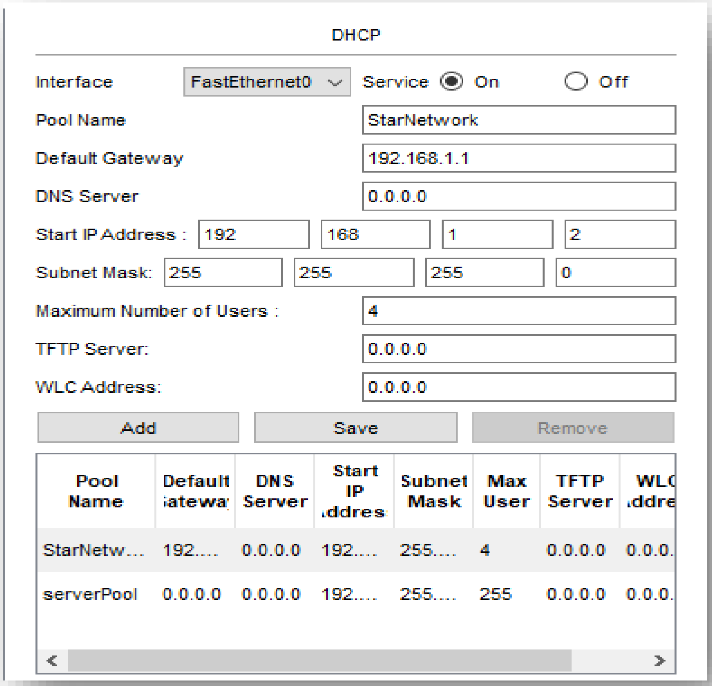
Assigning ip address



Enabling DHCP Service

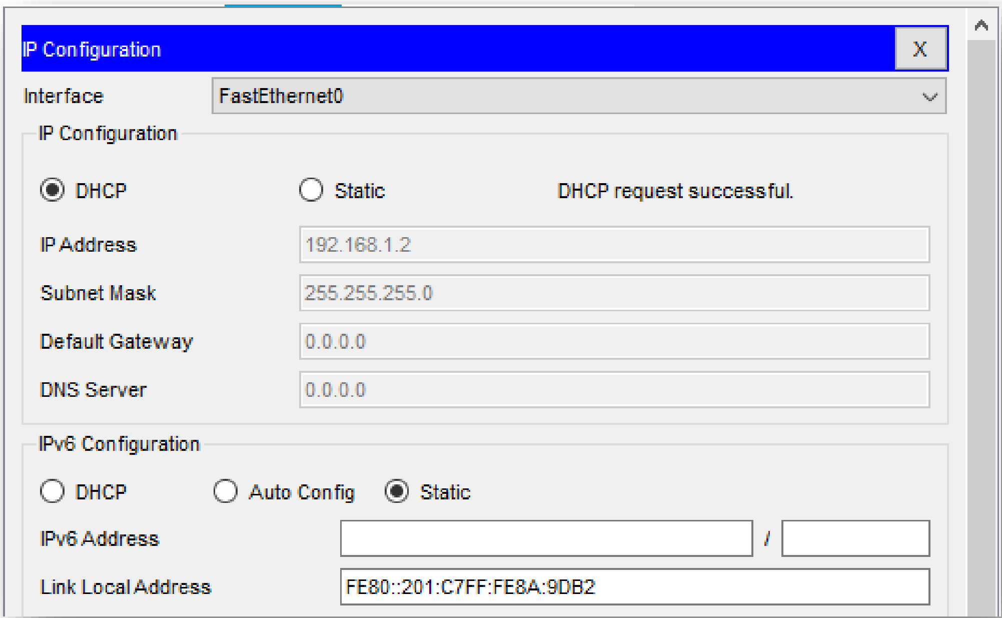


Configuring A DHCP Pool

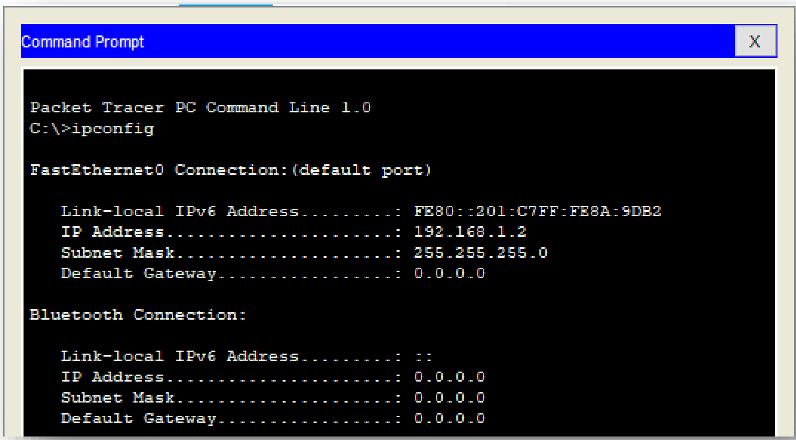


Adding a DHCP pool for the network, e.g. StarNetwork
Default Gateway: 192.168.1.1 (Server's IP)
Start IP Address: 192.168.1.2
Subnet Mask: 255.255.255.0
Max Users: Setting the number based on devices (e.g., 4 for 4 PCs).

Configuring the PCs



Command Prompt PC to ping the server
Verification



Steps for Wireshark

Step 1: Set Up DHCP on the Server
- Configure the server to act as a DHCP server as described previously (with the appropriate IP pool, gateway, and subnet).

Step 2: Start Wireshark to Capture DHCP Traffic
- Launch Wireshark on the machine connected to the same network.
- Select the appropriate network interface to capture traffic (e.g., Ethernet, Wi-Fi).
- In the Wireshark filter bar, type dhcp to filter only DHCP-related packets.

Step 3: Monitor DHCP Discovery (Client Request)
- On any client machine (PC), configure the IP to DHCP in the IP Configuration.
- As the PC requests an IP address, Wireshark will capture the DHCP Discover message.
  - Look for DHCP Discover packets from the client (PC).
  - These packets are broadcast requests from the client asking for a DHCP server.

Step 4: Monitor DHCP Offer (Server Response)
- The DHCP server responds to the client's Discover message with a DHCP Offer.
  - The server will send a DHCP Offer message, including the offered IP address, subnet mask, default gateway, and lease time.
  - In Wireshark, the DHCP Offer packet can be identified by its message type (DHCP Offer).

Step 5: Monitor DHCP Request (Client Confirmation)
- The client sends a DHCP Request message to the server, requesting to accept the offered IP address.
  - The DHCP Request packet will confirm the client's acceptance of the IP address provided by the DHCP server.

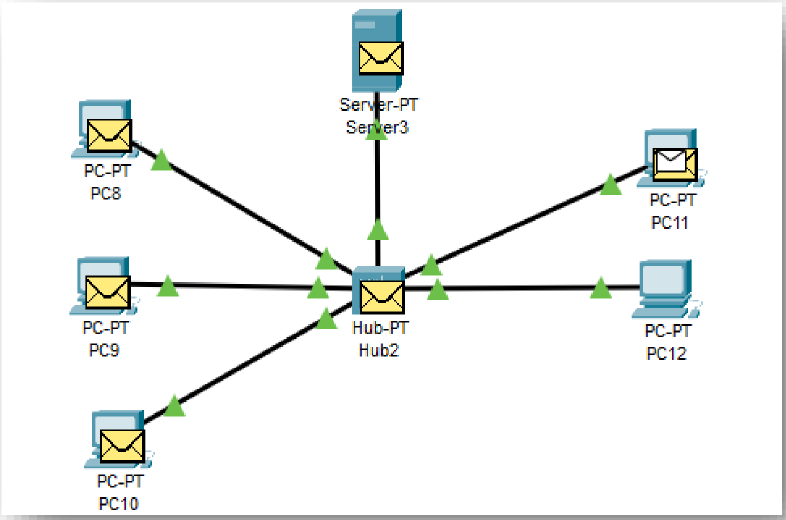
Step 6: Monitor DHCP Acknowledgment (Server Confirmation)
- The server responds with a DHCP Acknowledgment (ACK) message to confirm that the client has been assigned the IP address.
  - The DHCP ACK packet contains the IP address and other configuration details assigned to the client.

Step 7: Verify the Client's Assigned IP Address

- After completing the DHCP exchange, use ipconfig (Windows) on the client machine to confirm that it has received an IP address from the DHCP server.

Step 8: Analyze DHCP Leases and Renewals

- If the client's IP lease is near expiration, it will send a DHCP Request to renew the lease.
    - In Wireshark, look for the DHCP Request and DHCP ACK packets associated with lease renewal.



Simulation Panel

Event List

Vis.	Time(sec)	Last Device	At Device	Type
	5349.172	--	Server3	ICMP
	5349.172	--	Server3	ARP
	5349.673	Server3	Hub2	DHCP
	5349.674	Hub2	PC10	DHCP
	5349.674	Hub2	PC9	DHCP
	5349.674	Hub2	PC8	DHCP
	5349.674	Hub2	PC11	DHCP
	5349.674	Hub2	PC12	DHCP
	5349.675	PC12	Hub2	DHCP
	5349.676	Hub2	PC10	DHCP
	5349.676	Hub2	PC9	DHCP
	5349.676	Hub2	PC8	DHCP
	5349.676	Hub2	Server3	DHCP
	5349.676	Hub2	PC11	DHCP
	5349.677	Server3	Hub2	DHCP

DHCP Discover

- **Purpose**: The DHCP Discover message is the first step in the DHCP handshake, initiated by the client device to locate a DHCP server on the network.
- **How it works**:
  - When a device (e.g., a computer or a smartphone) is powered on, it does not have an IP address and sends out a DHCP Discover message as a broadcast to the network. This message is sent to the DHCP server to request an IP address.
  - The DHCP Discover packet contains the client's MAC address, which helps the server identify the client device.
  - The client broadcasts the request because it doesn't know the IP address of any DHCP server on the network.
- **Packet Details**:
  - The DHCP Discover message is a Broadcast (UDP port 67) that is sent to the network to find a DHCP server.
  - It typically includes information like the client's MAC address, transaction ID, and optionally, a list of parameters requested by the client (such as DNS server or subnet mask).

No.	Time	Source	Destination	Protocol	Lengtl	Info
1	0.000000	0.0.0.0	255.255.255.255	DHCP	314	DHCP Discover - Transaction ID 0x3d1d
2	0.000295	192.168.0.1	192.168.0.10	DHCP	342	DHCP Offer - Transaction ID 0x3d1d
3	0.070031	0.0.0.0	255.255.255.255	DHCP	314	DHCP Request - Transaction ID 0x3d1e
4	0.070345	192.168.0.1	192.168.0.10	DHCP	342	DHCP ACK - Transaction ID 0x3d1e

```
Dynamic Host Configuration Protocol (Discover)
    Message type: Boot Request (1)
    Hardware type: Ethernet (0x01)
    Hardware address length: 6
    Hops: 0
    Transaction ID: 0x00003d1d
    Seconds elapsed: 0
  > Bootp flags: 0x0000 (Unicast)
    Client IP address: 0.0.0.0
    Your (client) IP address: 0.0.0.0
    Next server IP address: 0.0.0.0
    Relay agent IP address: 0.0.0.0
    Client MAC address: GrandstreamN_01:fc:42 (00:0b:82:01:fc:42)
    Client hardware address padding: 00000000000000000000
    Server host name not given
    Boot file name not given
    Magic cookie: DHCP
  > Option: (53) DHCP Message Type (Discover)
  > Option: (61) Client identifier
  > Option: (50) Requested IP Address (0.0.0.0)
  > Option: (55) Parameter Request List
  > Option: (255) End
    Padding: 00000000000000
```

```
Option: (53) DHCP Message Type (Discover)
    Length: 1
    DHCP: Discover (1)
Option: (61) Client identifier
    Length: 7
    Hardware type: Ethernet (0x01)
    Client MAC address: GrandstreamN_01:fc:42 (00:0b:82:01:fc:42)
Option: (50) Requested IP Address (0.0.0.0)
    Length: 4
    Requested IP Address: 0.0.0.0
Option: (55) Parameter Request List
    Length: 4
    Parameter Request List Item: (1) Subnet Mask
    Parameter Request List Item: (3) Router
    Parameter Request List Item: (6) Domain Name Server
    Parameter Request List Item: (42) Network Time Protocol Servers
Option: (255) End
    Option End: 255
```

DHCP Offer

- **Purpose**: The DHCP Offer message is the server's response to the DHCP Discover message, providing the client with an available IP address and other configuration parameters.
- **How it works**:
  - The DHCP Offer is sent by the server as a Broadcast (UDP port 68) to the client that requested an IP address.
  - This message includes several key pieces of information:
    - **Offered IP address**: A valid IP address within the server's predefined DHCP pool.
    - **Subnet Mask**: Defines the network's subnet.
    - **Default Gateway**: The router IP to be used for communication with other networks.
    - **Lease Time**: The duration for which the offered IP address is valid.
    - **DHCP Server Identifier**: The IP address of the DHCP server.
  - The DHCP Offer helps the client decide which IP address to accept.
- **Packet Details**:
  - The DHCP Offer includes the offered IP address, subnet mask, default gateway, lease time, and other options like DNS servers if the DHCP server is configured to provide them.
  - This packet also includes the client's MAC address, helping the server track which device is requesting the IP.

No.	Time	Source	Destination	Protocol	Lengtl	Info
1	0.000000	0.0.0.0	255.255.255.255	DHCP	314	DHCP Discover - Transaction ID 0x
2	0.000295	192.168.0.1	192.168.0.10	DHCP	342	DHCP Offer - Transaction ID 0x
3	0.070031	0.0.0.0	255.255.255.255	DHCP	314	DHCP Request - Transaction ID 0x
4	0.070345	192.168.0.1	192.168.0.10	DHCP	342	DHCP ACK - Transaction ID 0x

```
Dynamic Host Configuration Protocol (Offer)
    Message type: Boot Reply (2)
    Hardware type: Ethernet (0x01)
    Hardware address length: 6
    Hops: 0
    Transaction ID: 0x00003d1d
    Seconds elapsed: 0
  > Bootp flags: 0x0000 (Unicast)
    Client IP address: 0.0.0.0
    Your (client) IP address: 192.168.0.10
    Next server IP address: 192.168.0.1
    Relay agent IP address: 0.0.0.0
    Client MAC address: GrandstreamN_01:fc:42 (00:0b:82:01:fc:42)
    Client hardware address padding: 00000000000000000000
    Server host name not given
    Boot file name not given
    Magic cookie: DHCP
  > Option: (53) DHCP Message Type (Offer)
  > Option: (1) Subnet Mask (255.255.255.0)
  > Option: (58) Renewal Time Value
  > Option: (59) Rebinding Time Value
  > Option: (51) IP Address Lease Time
  > Option: (54) DHCP Server Identifier (192.168.0.1)
  > Option: (255) End
    Padding: 00000000000000000000000000000000000000000000000000
```

```
Option: (53) DHCP Message Type (Offer)
    Length: 1
    DHCP: Offer (2)
Option: (1) Subnet Mask (255.255.255.0)
    Length: 4
    Subnet Mask: 255.255.255.0
Option: (58) Renewal Time Value
    Length: 4
    Renewal Time Value: 30 minutes (1800)
Option: (59) Rebinding Time Value
    Length: 4
    Rebinding Time Value: 52 minutes, 30 seconds (3150)
Option: (51) IP Address Lease Time
    Length: 4
    IP Address Lease Time: 1 hour (3600)
Option: (54) DHCP Server Identifier (192.168.0.1)
    Length: 4
    DHCP Server Identifier: 192.168.0.1
Option: (255) End
    Option End: 255
```

DHCP Request

- **Purpose**: The DHCP Request message is sent by the client to accept the IP address offered by the server in the DHCP Offer message.
- **How it works**:
    - Once the client receives one or more DHCP Offer messages, it selects one offer and sends a DHCP Request to confirm the IP address assignment to the server.
    - The client may send a DHCP Request to multiple DHCP servers, but it will accept only one DHCP Offer.
    - The DHCP Request message also serves as a notification to other DHCP servers that the client has accepted an offer from one server.
- **Packet Details**:
    - The DHCP Request message is a Broadcast (UDP port 68) to inform the network about the client's IP choice.
    - It includes the IP address being requested and the DHCP Server Identifier (the server that made the offer the client is accepting).
    - If the client is renewing its lease, this message can be used to confirm the continuation of an IP lease.

No.	Time	Source	Destination	Protocol	Lengtl	Info
1	0.000000	0.0.0.0	255.255.255.255	DHCP	314	DHCP Discover - Transaction ID 0x3
2	0.000295	192.168.0.1	192.168.0.10	DHCP	342	DHCP Offer - Transaction ID 0x3
3	0.070031	0.0.0.0	255.255.255.255	DHCP	314	DHCP Request - Transaction ID 0x3
4	0.070345	192.168.0.1	192.168.0.10	DHCP	342	DHCP ACK - Transaction ID 0x3

```
v Dynamic Host Configuration Protocol (Request)
    Message type: Boot Request (1)
    Hardware type: Ethernet (0x01)
    Hardware address length: 6
    Hops: 0
    Transaction ID: 0x00003d1e
    Seconds elapsed: 0
  > Bootp flags: 0x0000 (Unicast)
    Client IP address: 0.0.0.0
    Your (client) IP address: 0.0.0.0
    Next server IP address: 0.0.0.0
    Relay agent IP address: 0.0.0.0
    Client MAC address: GrandstreamN_01:fc:42 (00:0b:82:01:fc:42)
    Client hardware address padding: 00000000000000000000
    Server host name not given
    Boot file name not given
    Magic cookie: DHCP
  > Option: (53) DHCP Message Type (Request)
  > Option: (61) Client identifier
  > Option: (50) Requested IP Address (192.168.0.10)
  > Option: (54) DHCP Server Identifier (192.168.0.1)
  > Option: (55) Parameter Request List
  > Option: (255) End
    Padding: 00
```

```
v Option: (53) DHCP Message Type (Request)
    Length: 1
    DHCP: Request (3)
v Option: (61) Client identifier
    Length: 7
    Hardware type: Ethernet (0x01)
    Client MAC address: GrandstreamN_01:fc:42 (00:0b:82:01:fc:42)
v Option: (50) Requested IP Address (192.168.0.10)
    Length: 4
    Requested IP Address: 192.168.0.10
v Option: (54) DHCP Server Identifier (192.168.0.1)
    Length: 4
    DHCP Server Identifier: 192.168.0.1
v Option: (55) Parameter Request List
    Length: 4
    Parameter Request List Item: (1) Subnet Mask
    Parameter Request List Item: (3) Router
    Parameter Request List Item: (6) Domain Name Server
    Parameter Request List Item: (42) Network Time Protocol Servers
v Option: (255) End
    Option End: 255
```

DHCP Acknowledgment (ACK)

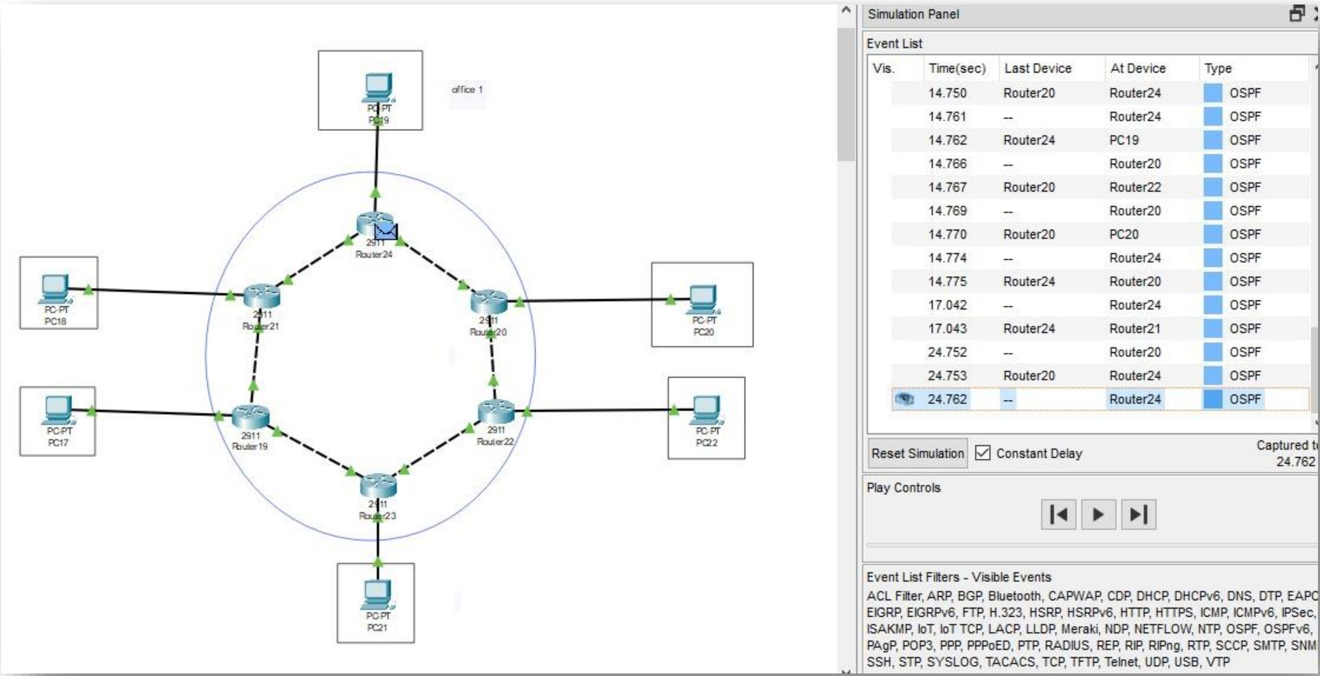
- **Purpose**: The DHCP Acknowledgment (ACK**)** message is the server's final confirmation to the client that the IP address has been successfully assigned, and the client can start using it.
- **How it works**:
  - Upon receiving the **DHCP Request, the DHCP server sends a DHCP ACK message to the client to confirm that the requested IP address has been assigned and is now valid.**
  - This message marks the completion of the DHCP process.
  - The client can now configure its network settings (IP address, subnet mask, gateway, DNS, etc.) and start communicating on the network.
- **Packet Details**:
  - The DHCP ACK includes the final confirmation of the IP address and other settings (such as lease time).
  - The message also marks the beginning of the client's lease period, during which the client can use the IP address.

No.	Time	Source	Destination	Protocol	Lengtl	Info
1	0.000000	0.0.0.0	255.255.255.255	DHCP	314	DHCP Discover - Transaction ID 0x3d1d
2	0.000295	192.168.0.1	192.168.0.10	DHCP	342	DHCP Offer - Transaction ID 0x3d1d
3	0.070031	0.0.0.0	255.255.255.255	DHCP	314	DHCP Request - Transaction ID 0x3d1e
4	0.070345	192.168.0.1	192.168.0.10	DHCP	342	DHCP ACK - Transaction ID 0x3d1e

```
Dynamic Host Configuration Protocol (ACK)
    Message type: Boot Reply (2)
    Hardware type: Ethernet (0x01)
    Hardware address length: 6
    Hops: 0
    Transaction ID: 0x00003d1e
    Seconds elapsed: 0
  > Bootp flags: 0x0000 (Unicast)
    Client IP address: 0.0.0.0
    Your (client) IP address: 192.168.0.10
    Next server IP address: 0.0.0.0
    Relay agent IP address: 0.0.0.0
    Client MAC address: GrandstreamN_01:fc:42 (00:0b:82:01:fc:42)
    Client hardware address padding: 00000000000000000000
    Server host name not given
    Boot file name not given
    Magic cookie: DHCP
  > Option: (53) DHCP Message Type (ACK)
  > Option: (58) Renewal Time Value
  > Option: (59) Rebinding Time Value
  > Option: (51) IP Address Lease Time
  > Option: (54) DHCP Server Identifier (192.168.0.1)
  > Option: (1) Subnet Mask (255.255.255.0)
  > Option: (255) End
    Padding: 0000000000000000000000000000000000000000000000000000
```

```
v Option: (53) DHCP Message Type (ACK)
    Length: 1
    DHCP: ACK (5)
v Option: (58) Renewal Time Value
    Length: 4
    Renewal Time Value: 30 minutes (1800)
v Option: (59) Rebinding Time Value
    Length: 4
    Rebinding Time Value: 52 minutes, 30 seconds (3150)
v Option: (51) IP Address Lease Time
    Length: 4
    IP Address Lease Time: 1 hour (3600)
v Option: (54) DHCP Server Identifier (192.168.0.1)
    Length: 4
    DHCP Server Identifier: 192.168.0.1
v Option: (1) Subnet Mask (255.255.255.0)
    Length: 4
    Subnet Mask: 255.255.255.0
v Option: (255) End
    Option End: 255
```

Dynamic Routing Using Packet Tracer (OSPF)



Step 1: Network Subnet Allocation

- **Router-to-Router Links**
  - R1 ↔ R2: 192.168.1.0/30
  - R2 ↔ R3: 192.168.2.0/30
  - R3 ↔ R4: 192.168.3.0/30
  - R4 ↔ R5: 192.168.4.0/30
  - R5 ↔ R6: 192.168.5.0/30
  - R6 ↔ R1: 192.168.6.0/30

Example

```
Router# enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface G0/0
Router(config-if)#ip address 192.168.2.2 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface G0/1
Router(config-if)#ip address 192.168.3.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
Router#
```

- **Router-to-PC Links**
  - R1 ↔ PC1: 10.0.1.0/24
  - R2 ↔ PC2: 10.0.2.0/24
  - R3 ↔ PC3: 10.0.3.0/24
  - R4 ↔ PC4: 10.0.4.0/24
  - R5 ↔ PC5: 10.0.5.0/24
  - R6 ↔ PC6: 10.0.6.0/24

Example

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface G0/2
Router(config-if)#ip address 10.0.3.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
```

Step 2: Router Configuration

Each router must be configured to assign IP addresses to interfaces and enable OSPF. Below are the CLI commands for each router.

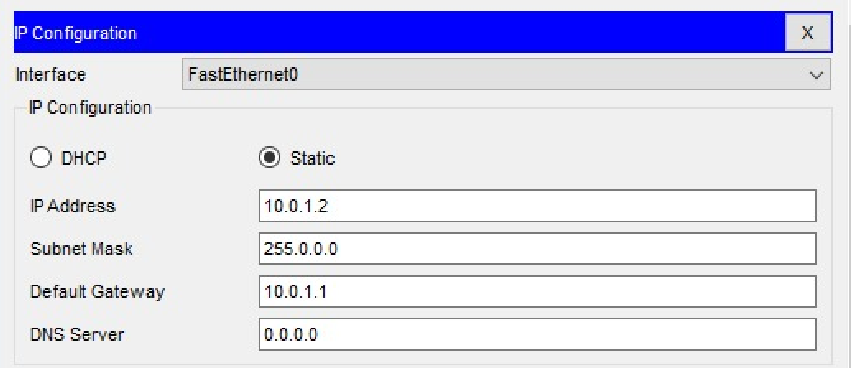
```
Router> enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 192.168.1.0 0.0.0.3 area 0
Router(config-router)#network 192.168.6.0 0.0.0.3 area 0
Router(config-router)# network 10.0.1.0 0.0.0.255 area 0
Router(config-router)#exit
Router(config)#exit
```

Step 3: Verification

```
Router>show ip interface brief
Interface              IP-Address      OK? Method Status
Protocol
GigabitEthernet0/0     192.168.1.1     YES manual up
up
GigabitEthernet0/1     192.168.6.1     YES manual up
up
GigabitEthernet0/2     10.0.1.1        YES manual up
up
Vlan1                  unassigned      YES unset  administratively
down down
Router>show ip interface brief
Interface              IP-Address      OK? Method Status
Protocol
GigabitEthernet0/0     192.168.1.1     YES manual up
up
GigabitEthernet0/1     192.168.6.1     YES manual up
up
GigabitEthernet0/2     10.0.1.1        YES manual up
up
Vlan1                  unassigned      YES unset  administratively
down down
```

Step 4: PC Configuration

Each PC is configured with a static IP address, subnet mask, and default gateway matching its connected router. Below are the configurations for each PC:



Step 5: Verifying ip In Pc

```
C:\>ping 10.0.1.1

Pinging 10.0.1.1 with 32 bytes of data:

Reply from 10.0.1.1: bytes=32 time=1ms TTL=255
Reply from 10.0.1.1: bytes=32 time<1ms TTL=255
Reply from 10.0.1.1: bytes=32 time<1ms TTL=255
Reply from 10.0.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 10.0.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Step 6: Verify OSPF Configuration

- Check OSPF neighbors
- View OSPF routes

```
Router#show ip ospf neighbor

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B -
BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS
inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.1.0/24 is directly connected, GigabitEthernet0/2
L       10.0.1.1/32 is directly connected, GigabitEthernet0/2
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/30 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
     192.168.6.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.6.0/30 is directly connected, GigabitEthernet0/1
L       192.168.6.1/32 is directly connected, GigabitEthernet0/1
```

- View OSPF database

```
Router#show ip ospf database
            OSPF Router with ID (192.168.6.1) (Process ID 1)

                Router Link States (Area 0)

Link ID         ADV Router      Age         Seq#       Checksum Link
count
192.168.6.1     192.168.6.1     301         0x80000003 0x0075fa 3
Router#
```

User Datagram protocol(UDP) using WireShark

Wireshark can be used to analyze UDP traffic by capturing and inspecting UDP packets between devices. UDP is a connectionless protocol that does not guarantee delivery or error-checking, so it's ideal for real-time applications like video or voice communication.

Steps to analyze UDP using Wireshark:

Step 1: Start Capturing Traffic

- Open Wireshark and start a capture session on the network interface you want to monitor.

Step 2: Filter UDP Traffic

- Apply a display filter to capture only UDP traffic by typing udp in the filter bar.

Step 3: Examine UDP Packets

- Look at the source and destination IP addresses, source and destination ports, and payload data within the captured UDP packets.

Step 4: Analyze Packet Details

- Inspect the UDP header, which includes the source and destination ports, length, and checksum.

Step 5: Monitor UDP Communication

- Use the "Statistics" menu in Wireshark to analyze the overall UDP traffic and observe patterns like packet loss or delays in real-time communication.

Packet Filtering UDP

udp

No.	Time	Source	Destination	Protocol	Lengtl	Info
1051	19:03:47.497313	2407:d000:a:2f3c:c2…	2001:4860:4864:5:40…	UDP	258	51981 → 3478 Len=196
1052	19:03:47.506468	2407:d000:a:2f3c:c2…	2001:4860:4864:5:40…	RTCP	138	Sender Report
1053	19:03:47.509975	2407:d000:a:2f3c:c2…	2001:4860:4864:5:40…	UDP	1236	55947 → 3478 Len=1174
1054	19:03:47.510532	2407:d000:a:2f3c:c2…	2001:4860:4864:5:40…	UDP	1236	55947 → 3478 Len=1174
1055	19:03:47.510733	2407:d000:a:2f3c:c2…	2001:4860:4864:5:40…	UDP	1236	55947 → 3478 Len=1174
1056	19:03:47.510853	2407:d000:a:2f3c:c2…	2001:4860:4864:5:40…	UDP	1236	55947 → 3478 Len=1174
1057	19:03:47.510920	2407:d000:a:2f3c:c2…	2001:4860:4864:5:40…	UDP	1236	55947 → 3478 Len=1174
1058	19:03:47.510991	2407:d000:a:2f3c:c2…	2001:4860:4864:5:40…	UDP	1236	55947 → 3478 Len=1174
1059	19:03:47.511076	2407:d000:a:2f3c:c2…	2001:4860:4864:5:40…	UDP	1236	55947 → 3478 Len=1174
1060	19:03:47.511135	2407:d000:a:2f3c:c2…	2001:4860:4864:5:40…	UDP	1236	55947 → 3478 Len=1174
1061	19:03:47.511213	2407:d000:a:2f3c:c2…	2001:4860:4864:5:40…	UDP	1236	55947 → 3478 Len=1174
1062	19:03:47.511272	2407:d000:a:2f3c:c2…	2001:4860:4864:5:40…	UDP	1237	55947 → 3478 Len=1175

Detail of UDP

```
User Datagram Protocol, Src Port: 55947, Dst Port: 3478
    Source Port: 55947
    Destination Port: 3478
    Length: 1182
    Checksum: 0xe4dd [unverified]
    [Checksum Status: Unverified]
    [Stream index: 0]
    [Stream Packet Number: 895]
    [Timestamps]
        [Time since first frame: 2.816353000 seconds]
        [Time since previous frame: 0.003507000 seconds]
    UDP payload (1174 bytes)
```

Conclusion

In conclusion, this project demonstrates the effective implementation of a star topology network, highlighting its centralized structure, scalability, and ease of management. By integrating key protocols such as IP, ICMP, SSL, VLAN, DHCP, OSPF, and UDP, it showcases robust communication, secure data transmission, and efficient network performance. Tools like Wireshark enhanced the analysis, offering practical insights into data flow and troubleshooting. This simulation not only reinforces the practicality of star topology but also emphasizes its adaptability for modern networking needs, providing a solid foundation for further exploration of advanced network designs.